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SUMMARY KEYWORDS

model components, minimal assumptions, data fitting, maximum likelihood, least squares, generalizable model, system level, normalized average, Gaussian variables, analytical equation, derivative setting, curve fitting, deliverable, focused work, dental appointment

SPEAKERS

Alex Gates, Charles Levine, Speaker 1

Charles Levine 00:03
Okay, first part, you want me to do…

Alex Gates 00:07
yeah,

Charles Levine 00:07
articulate to you my model, how I'm planning to build it. What are the components of it? What

Alex Gates 00:14
are the minimal assumptions? The

Charles Levine 00:16
minimal assumptions are variables, right,

Alex Gates 00:19
right, exactly. And then the second part is, what's the game plan for fitting that to the data? So either you fit some of it already, you've managed to work it into the same framework, or you see how what this is in the model corresponds to these features of the data that you can easily fit. And then you need to have a plan for how that whole thing is going to get fit. So, is it a maximum likelihood estimator? Is it a minimum square, least squares estimator?

Charles Levine 00:49
You're talking about fit. You're saying, hey, you've got this master equation. What are you going to do to tweak the variables so that it...

Alex Gates 00:56
represents the systems? Okay? Right? Yeah. So ultimately, what you need is, you know, in some respects, it should just be a generalizable model function where you're going to plug in data, and it's

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going to fit all the pieces of the model together out of the data, right? And then that will return to you that this, you know, again, I don't know the level of the unit that you're modeling. So, are you modeling an individual? Are you modeling a unit? Are you modeling the system? And it's a system level. So again, these are where you have to kind of articulate, what are the different pieces? What are we trying to hit?

Charles Levine 01:35
Right?

Alex Gates 01:35
This then corresponds to this average, or this normalized average, divided by what's in the pool, or divided by what's in the whole population, or, you know, you have to kind of really, these variables have to be between zero and one, otherwise they don't make sense.

Charles Levine 01:51
Check

Alex Gates 01:52
these other ones have to be Gaussian, right? And we're going to make that standardizing assumption, and it's okay, you know, because our data looks

Charles Levine 02:00
okay,

Alex Gates 02:01
right? Like, yeah, these are all the pieces that we're we have to really think,

Speaker 1 02:04
yeah,

Alex Gates 02:05
right,

Charles Levine 02:05
yep.

Alex Gates 02:06
And so what I want to see is that at least the beginning or the plan for how you would do that,

Charles Levine 02:11
okay,

Alex Gates 02:12
right?

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Charles Levine 02:13
Yep.

Alex Gates 02:14
That's, that's where we are. And I think today, after you've worked on it for this long two days, more to do that is not unreasonable, but it's a focused two days trying to get that deliverable.

Charles Levine 02:26
Yeah,

Alex Gates 02:27
prepared.

Charles Levine 02:28
Okay, sounds good. I have a dental appointment at 2:30, but I'll be back. It occurred to me that if I should, if I can somehow create an analytical equation for the the derivative setting the derivative of that curve equal to zero.

Alex Gates 02:49
That's one way to fit those models.

Charles Levine 02:50
That is when you stop and create, that's when you know, you've maximized, you know,

Alex Gates 02:56
yep. So awesome. Exactly. That's, that's what we end up wanting to be able to pull out.

Speaker 1 03:01
Yeah, that's

Alex Gates 03:02
actually what we want to pull

Charles Levine 03:04
out. Okay, we'll kill

Alex Gates 03:05
all right. Power through. I.

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